

HOME AUTOMATION Using Android

KARTHIK RAJASEKARAN

Nowadays, people have smartphones with them all the time. So it makes sense to use these to control home appliances. Presented here is a home automation system using a simple Android app, which you can use to control electrical appliances with clicks or voice commands. Commands are sent via Bluetooth to Arduino Uno. So you need not get up to switch on or switch off the device while watching a movie or doing some work.

Circuit and working

The circuit is built around Arduino Uno board, Bluetooth module HC-05 and a 3-channel relay board. The number of channels depends on the number of appliances you wish to control. Arduino Uno is powered with a 12V DC adaptor/ power source. The relay module and Bluetooth module can be, in turn, powered using a board power supply of Arduino Uno. Author's prototype is shown in Fig. 1. Con-

nection details for each appliance are shown in Fig. 2.

Bluetooth module. Bluetooth module used in this project is HC-05 (Fig. 4), which supports master and slave mode serial communication (9600-115200 bps) SPP and UART interface. Using these features it can communicate with other Bluetooth-enabled devices like mobile phones, tablets and laptops. The module runs on 3.3V to 5V power supply.

Relay module. A relay allows you to turn on or turn off a circuit using

voltage and/or current much higher than what Arduino could handle. Relay provides complete isolation between the low-voltage circuit on Arduino side and the high-voltage side controlling the load. It gets activated using 5V from Arduino, which, in turn, controls electrical appliances like fans, lights and air-conditioners. An 8-channel relay module is shown in Fig. 5.

Arduino Uno board. Arduino is an open source electronics prototyping platform based on flexible, easy-to-use hardware and software. It is intended for artists, designers, hobbyists and anyone interested in creating interactive objects or environments.

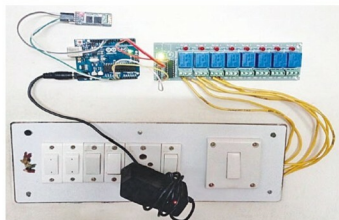


Fig. 1: Author's prototype

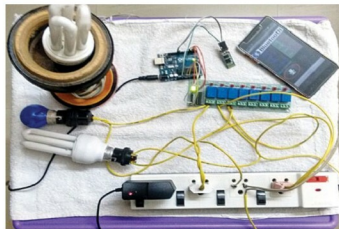


Fig. 2: Connections for the appliances

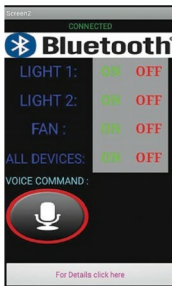


Fig. 3: Control panel on Android smartphone



Fig. 4: Bluetooth module